

Weekly Assessment – AI Generation & Validation (Detailed Documentation)

1. Introduction

This document explains the complete process of building an AI-based weekly assessment system. The system automatically generates questions, validates answers using AI (RAG), and provides scoring without manual intervention.

2. Concept

The system uses Artificial Intelligence to generate weekly assessments based on the videos watched by trainees. It evaluates answers intelligently and provides instant results.

3. Workflow

Step 1: Collect weekly learning data (video scripts/content).

Step 2: AI generates questions based on the content.

Step 3: Trainee submits answers.

Step 4: AI validates answers using RAG (content-based checking).

Step 5: System calculates score and displays result.

4. AI Technologies Used

1. AI for Question Generation:

Tools: OpenAI (GPT), Google Gemini

Function: Generates MCQs, True/False, and subjective questions from learning content.

Feature: Automatically creates unique questions every week.

2. AI for Answer Validation (RAG):

Tools: OpenAI / Gemini + Stored Content

Function: Compares user answers with actual learning material.

Feature: Ensures answers are validated based on real content, not general AI knowledge.

3. AI for Semantic Scoring:

Tools: OpenAI GPT / Gemini

Function: Evaluates subjective answers based on meaning and logic.

Feature: Understands intent instead of matching keywords.

5. What is RAG?

RAG (Retrieval Augmented Generation) is a technique where AI retrieves relevant data from stored content and uses it to generate accurate answers or validations. It ensures the system uses internal learning material.

6. Scoring System

Objective Questions: Auto-scored using correct answers.

Subjective Questions: AI gives score based on understanding and correctness.

7. Free vs Paid Implementation

Free Version

Tools Used:

AI Model: Google Gemini (Free Tier)

RAG (Basic): Manual content passing (No extra tool)

Optional Vector Search: FAISS (Free, open-source)

Features:

- AI generates questions using Gemini API (limited usage)
- Content is stored manually in database
- Validation is done by sending:

Paid Version

🔗 Tools Used:

- **AI Model:** OpenAI GPT-4 / GPT-4o
- **Vector Database (RAG):** Pinecone / Weaviate
- **Backend:** Node.js

Features:

- High-quality AI question generation
- Advanced RAG:
 - Searches relevant content automatically
 - Gives very accurate validation
- Smart semantic scoring (understands logic, not just keywords)
- Fast and scalable system
- Fully automated workflow

8. How to Start

Step 1: Store learning content in database.

Step 2: Use AI to generate questions.

Step 3: Build simple UI for answers.

Step 4: Validate answers using AI.

Step 5: Add scoring logic.

9. Benefits

- No manual grading
- Fast and scalable
- Accurate evaluation
- Personalized assessments